

Systematic Artifact Analysis and Null Hypothesis Stress Testing for Emergent Vacuum Response Experiments

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Abstract

This work presents a systematic analysis of potential experimental artifacts and false-positive mechanisms relevant to testing Emergent Vacuum Response Theory (EVRT). Building on the constrained phenomenological framework introduced in prior work, the goal of this study is to rigorously examine how conventional physical effects may mimic or obscure small residual forces in nonequilibrium electromagnetic systems.

Rather than attempting to confirm the existence of emergent vacuum responses, this paper emphasizes falsification pathways and null hypothesis stress testing. Known sources of experimental error—including thermal gradients, mechanical vibration, electromagnetic coupling, electrohydrodynamic effects, and measurement bias—are analyzed in terms of their capacity to produce spurious force signals.

The paper defines explicit discrimination criteria, control strategies, and failure modes to distinguish genuine signals from artifacts. This approach ensures that any claimed non-null result must survive increasingly stringent experimental constraints, while null results contribute directly to tightening bounds on the effective response parameter χ_{eff} .

The objective is to establish a disciplined methodology for evaluating EVRT-related experiments, reinforcing the framework’s commitment to falsifiability, reproducibility, and consistency with established physics.

1 Introduction

The Emergent Vacuum Response Theory (EVRT) framework introduces a constrained phenomenological parameterization for investigating whether structured, nonequilibrium electromagnetic systems may exhibit small residual stress-energy effects beyond conventional expectations. As emphasized in the capstone framework, the interpretation of any observed signal requires rigorous exclusion of conventional physical mechanisms.

The present work addresses this requirement directly by systematically analyzing potential sources of experimental artifacts that could produce false-positive signals. The objective is not to demonstrate the existence of EVRT effects, but to define the conditions under which such effects could be meaningfully distinguished from known physical phenomena.

2 Null Hypothesis Framework

The baseline null hypothesis assumes:

- $\chi_{\text{eff}} = 0$, such that all system behavior is fully described by standard electrodynamics.

- Any measured force or asymmetry arises from known physical mechanisms or experimental error.

Under this hypothesis, all observed signals must be explainable through conventional processes, including thermal, mechanical, electromagnetic, and environmental effects.

3 Classes of Experimental Artifacts

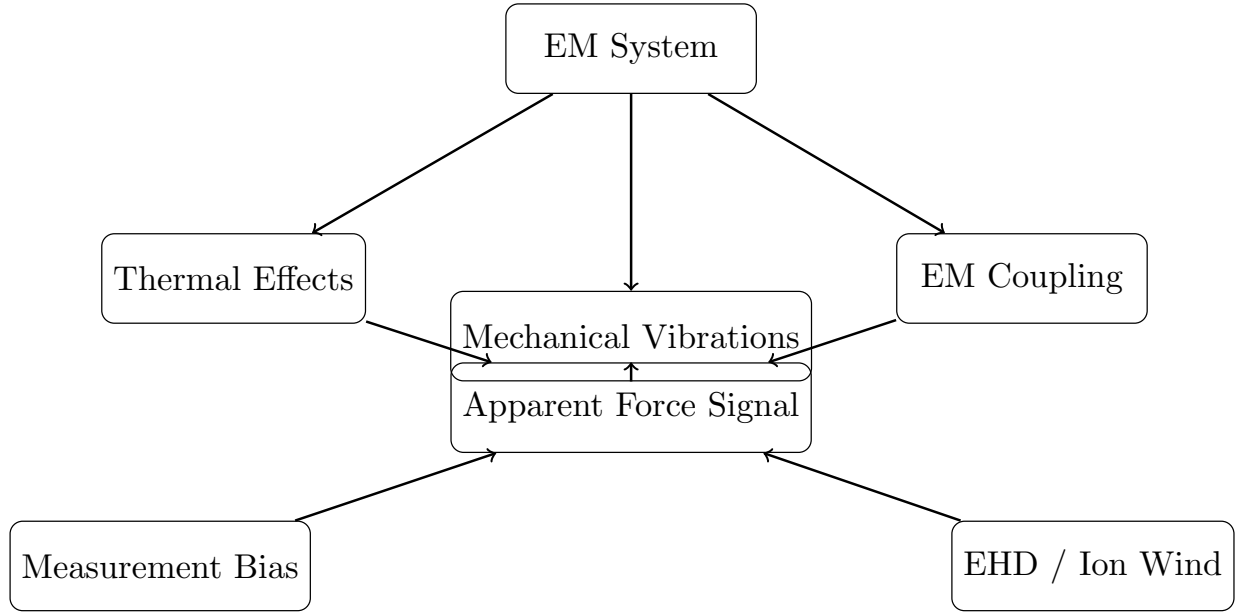


Figure 1: Systematic Artifact Analysis and Null Hypothesis Stress Testing for Emergent Vacuum Response Experiments

These artifact classes vary in severity and likelihood, with thermal and electrohydrodynamic effects typically dominating early-stage measurements, while electromagnetic coupling and measurement bias become more prominent at higher precision levels.

3.1 Thermal Effects

Temperature gradients can produce apparent forces through:

- asymmetric expansion of structural components
- convection currents in non-ideal vacuum conditions
- thermal radiation pressure imbalance

Thermal gradients are known to produce apparent forces through asymmetric expansion and radiative effects [7]. Even small temperature differences can generate forces comparable to nanonewton sensitivity levels.

3.2 Mechanical Vibrations

Vibrational coupling can introduce spurious signals through:

- environmental vibrations (building, acoustic noise)
- mechanical resonances within the apparatus
- coupling between cables and support structures

These effects may produce periodic or directional signals that mimic expected EVRT signatures.

3.3 Electromagnetic Coupling

Unintended electromagnetic interactions include:

- Lorentz forces on current-carrying conductors
- coupling to external electromagnetic fields
- ground loops and induced currents

Such effects can create directional forces correlated with system operation.

3.4 Electrohydrodynamic Effects

Ion wind and related electrohydrodynamic (EHD) phenomena can generate:

- net thrust in atmospheric or partial vacuum conditions
- asymmetric charge transport
- momentum transfer through ionized gas

Electrohydrodynamic thrust effects are well-documented in high-voltage systems [8].

3.5 Measurement Bias and Instrumentation

Measurement systems may introduce artifacts through:

- drift in sensors or balances
- electronic noise and offset errors
- data processing bias

Such effects can create persistent but non-physical signals.

4 Failure Modes and False Positives

Each artifact class can produce signals that satisfy some, but not all, EVRT prediction criteria:

- Thermal gradients can produce steady directional forces.
- Vibrations can generate repeatable but phase-dependent signals.
- Electromagnetic coupling can scale with power input.
- EHD effects can persist under certain environmental conditions.

Therefore, no single observation is sufficient to establish a non-null result.

5 Discrimination Criteria

To distinguish genuine effects from artifacts, a valid signal must:

- persist under vacuum conditions eliminating EHD effects
- remain after thermal stabilization
- be independent of mechanical coupling pathways
- survive changes in grounding and electromagnetic shielding

Failure of any of these criteria indicates a signal dominated by conventional physical effects.

6 Experimental Control Strategies

Key control approaches include:

- symmetry reversal tests
- power cycling with blind conditions
- environmental isolation (thermal and vibrational)
- electromagnetic shielding and grounding verification

These methods are designed to eliminate known sources of false positives.

7 Null Hypothesis Stress Testing

A rigorous experimental program should aim to:

- systematically eliminate each artifact class
- reduce measurement uncertainty below target thresholds
- repeat experiments under varied conditions

Only signals that persist through all such tests can be considered candidates for further investigation.

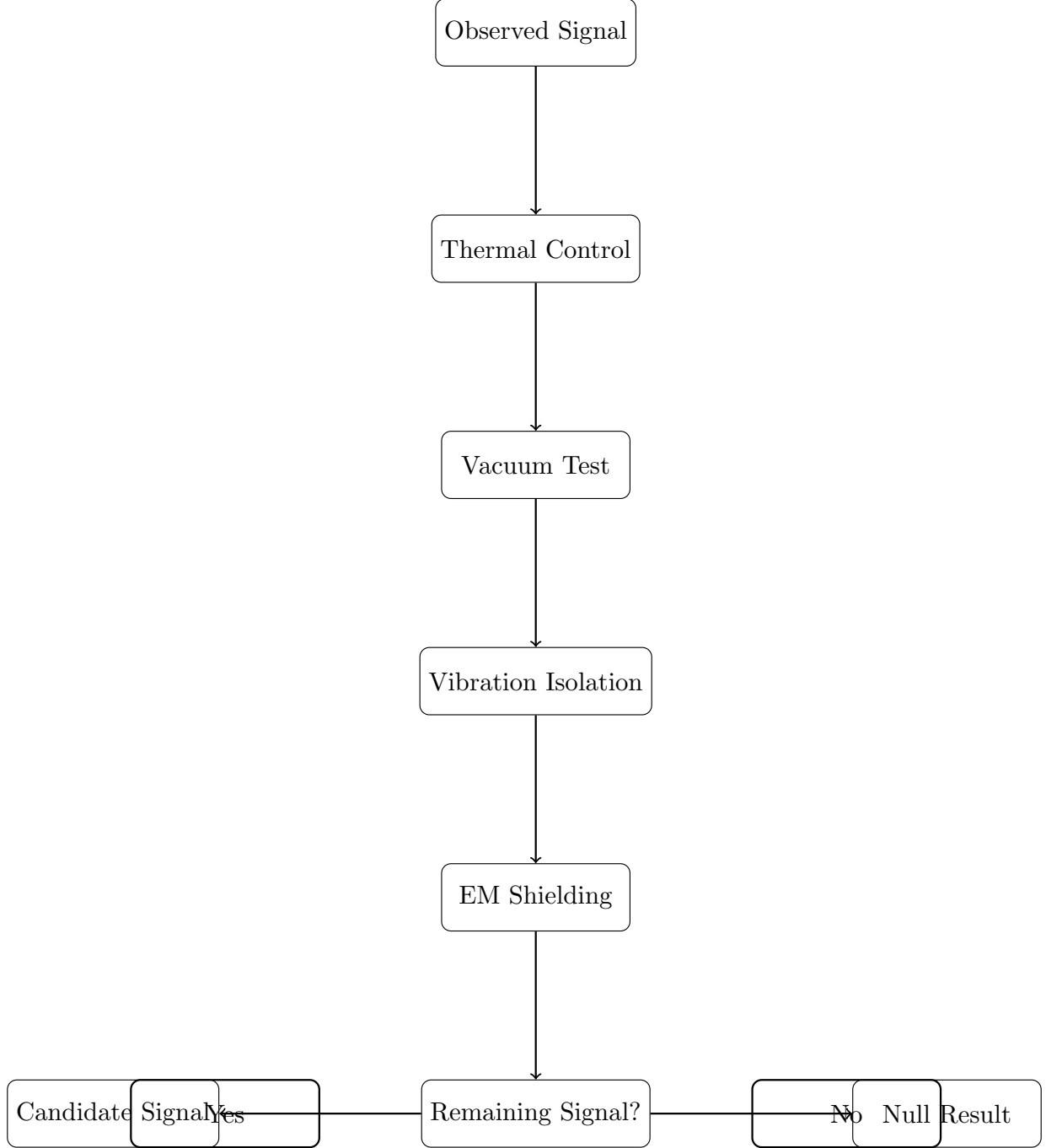


Figure 2: Sequential control strategy for null hypothesis stress testing. A valid signal must persist through all stages of artifact elimination to be considered a candidate for further investigation.

8 Implications for χ_{eff} Constraints

Null results under increasingly stringent conditions directly constrain the parameter space of χ_{eff} . As sensitivity improves, the allowable range of any emergent response shrinks accordingly.

9 Discussion

The majority of apparent signals in early-stage experiments are expected to be attributable to conventional physical effects rather than emergent phenomena. This analysis reinforces the central principle of the EVRT framework: that null results are scientifically valuable.

The majority of apparent signals in early-stage experiments are expected to be attributable to conventional physical effects rather than emergent phenomena. This analysis reinforces the central principle of the EVRT framework: that null results are scientifically valuable. This layered control structure emphasizes that most apparent signals are expected to be eliminated through successive artifact suppression stages.

The systematic identification and elimination of these effects is therefore essential not only for detecting potential new phenomena, but also for establishing meaningful experimental bounds.

10 Conclusion

This work establishes a structured methodology for identifying and eliminating experimental artifacts in EVRT-related investigations. By prioritizing falsification and null hypothesis stress testing, the framework ensures that any surviving signal must satisfy stringent physical and experimental criteria.

This approach strengthens the credibility of the broader EVRT program and provides a clear pathway for future experimental work grounded in established physics.

This work establishes that any credible detection of an EVRT-consistent signal must first survive systematic elimination of conventional physical effects, which are expected to dominate early experimental observations.

References

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